

Week 2: Data Basics

Day 2: Data Cleaning Fundamentals

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Abstract

Welcome to Day 2 of Week 2! Today, we transition from understanding data to actively preparing it for Machine Learning. Raw data is rarely perfect; it contains errors, duplicates, and gaps that can severely degrade AI model performance. This session focuses on the critical, hands-on skills of data cleaning using spreadsheet tools.

- **The Core Philosophy:** Understanding "Garbage In, Garbage Out" and the impact of messy data on ML accuracy.
- **Data Manipulation:** Mastering sorting and filtering to organize and isolate specific data subsets.
- **Deduplication:** Safely identifying and removing duplicate records without losing unique data.
- **Missing Values:** Strategies for handling blanks, "N/A", and inconsistent placeholders through deletion or imputation.

Session Objectives

By the end of today's session, learners will be able to:

- Explain why data cleaning is the most critical step before training an ML model.
- Sort data alphabetically, numerically, and by date to spot trends and outliers.
- Filter data to isolate specific subsets for targeted analysis.
- Safely identify and remove duplicate entries using built-in spreadsheet tools.
- Recognize various forms of missing data (blanks, "Null", "N/A", "-").
- Apply appropriate strategies (deletion or imputation) to handle missing values.

Part 1: Why Clean Data?

"Garbage In, Garbage Out – Preparing Data for ML"

The Golden Rule: Garbage In, Garbage Out

- **GIGO (Garbage In, Garbage Out):** A fundamental concept in computer science and AI.
- If you feed a Machine Learning model flawed, messy, or biased data, the model's predictions will be equally flawed.
- An ML model cannot distinguish between "true patterns" and "data entry errors." It simply learns whatever you give it.
- Therefore, data cleaning is not just administrative work; it is the foundation of model accuracy.

The Reality of AI

A highly complex, expensive AI model trained on dirty data will perform worse than a simple model trained on clean, high-quality data.

The Impact of Messy Data on ML Accuracy

- **Skewed Patterns:** Outliers or errors can pull the model's mathematical averages in the wrong direction.
- **Overfitting:** If the model learns specific data entry errors, it will fail to generalize to new, real-world data.
- **Algorithm Crashes:** Many ML algorithms will simply stop running or throw fatal errors if they encounter missing values or text in a numerical column.
- **Biased Decisions:** Duplicates can artificially inflate the importance of certain records, leading to biased predictions.

Real-World Consequences of Bad Data

Financial Services (Mobile Lending)

Duplicate customer records might make a user appear to have multiple active loans, leading to an unfair denial of credit.

Healthcare

Missing values in patient vitals could cause an ML diagnostic tool to miss a critical warning sign, leading to delayed treatment.

Retail & Supply Chain

Incorrect product categories (e.g., "Electronics" vs "Electonics") will fragment sales data, causing the AI to fail at inventory forecasting.

The Data Cleaning Pipeline

Today, we will execute the foundational steps of the data cleaning pipeline:

1. **Organize:** Sort and filter to understand the data's shape and spot obvious errors.
2. **Deduplicate:** Remove exact copies of records to prevent data inflation.
3. **Handle Missing Values:** Decide whether to drop incomplete records or fill in the gaps (imputation).
4. **Standardize:** Ensure consistent formatting (e.g., fixing "Nairobi", "nairobi", "NBO").

Note

We will perform Steps 1, 2, and 3 today using Excel and Google Sheets. Step 4 will be covered as we advance to Python next week.

Part 2: Sorting & Filtering

"Organizing and isolating data subsets."

What is Sorting?

- **Definition:** Arranging data in a specific, logical order based on the values in a selected column.
- **Common Sort Orders:**
  - **Alphabetical (A-Z / Z-A):** For text data (e.g., Customer Names, Cities).
  - **Numerical (Smallest to Largest / Largest to Smallest):** For quantities (e.g., Sales Amount, Age).
  - **Chronological (Oldest to Newest / Newest to Oldest):** For dates and times (e.g., Transaction Date).

Sorting is usually the very first step in Exploratory Data Analysis (EDA) because it makes patterns and anomalies immediately visible.

Why Sort Data?

- **Spotting Outliers:** Sorting a "Sales" column from highest to lowest instantly reveals extreme values (e.g., a sale of 1,000,000 KES when the average is 5,000 KES).
- **Finding Extremes:** Quickly identifying the minimum and maximum values in a dataset.
- **Grouping Categories:** Sorting by "Branch" brings all "Nairobi" records together, making it easier to spot missing entries or formatting inconsistencies.
- **Preparing for Filtering:** While not strictly required, sorting can make manual review of filtered subsets much faster.

How to Sort in Excel & Google Sheets

Step-by-Step Guide

1. Select the column header you want to sort by (e.g., "Age").
2. **In Excel:** Go to the `Data` tab → Click `Sort A to Z` or `Sort Z to A`.
3. **In Google Sheets:** Go to `Data` → `Sort sheet by column [Name] (A → Z)`.

Crucial Warning!

Always ensure you select the **entire dataset** or use the "Expand the selection" prompt. If you only sort one column, your data will become misaligned (e.g., Customer A's name will be matched with Customer B's age).

What is Filtering?

- **Definition:** Temporarily hiding rows that do not meet specific criteria, allowing you to focus only on a subset of the data.
- Unlike sorting, filtering does not change the order of the data; it simply hides the irrelevant rows from view.
- **Examples of Filter Criteria:**
  - Show only rows where "City" = "Nairobi".
  - Show only rows where "Age" $>$ 30.
  - Show only rows where "Status" = "Active".

Why Filter Data?

- **Isolate Specific Subsets:** If you are analyzing branch performance, filtering by "Nairobi" allows you to clean and review only that branch's data without being distracted by the rest.
- **Identify Missing Values:** You can filter a column to show only "Blanks", instantly revealing all rows with missing data.
- **Spot Inconsistencies:** Filtering a "City" column might reveal that you have "Nairobi", "nairobi", and "NBI". You can then manually correct them to a single standard.
- **Reduce Cognitive Load:** Working with 100 filtered rows is much easier than scrolling through 10,000 raw rows.

How to Filter in Excel & Google Sheets

Step-by-Step Guide

1. Click anywhere inside your dataset.
2. **In Excel:** Go to the `Data` tab → Click the `Filter` button (funnel icon). Dropdown arrows will appear on your headers.
3. **In Google Sheets:** Go to `Data` → `Create a filter`.
4. Click the dropdown arrow on a column header, uncheck "Select All", and check only the values you want to see (e.g., "Nairobi").

To remove the filter and see all data again, simply click the `Clear Filter` or turn off the filter toggle.

Sorting & Filtering: Best Practices

- **Always use Headers:** Ensure your first row contains clear column names. Sorting and filtering tools rely on headers to identify data boundaries.
- **Avoid Merged Cells:** Merged cells in the header row will break sorting and filtering. Keep headers strictly in a single, unmerged row.
- **Check Data Types:** If a numerical column is sorting alphabetically (e.g., 1, 10, 11, 2), it means the column is formatted as "Text". Change the format to "Number" first.
- **Clear Filters Before Saving:** If you share a filtered file, the recipient might not realize data is hidden. Always clear filters before exporting the final clean CSV.

Hands-on Exercise: Sorting & Filtering (Setup)

Preparation

- Open the "messy" sample dataset provided for Day 2 (`messy_sales_data.csv`).
- Ensure the file is properly separated into columns (use "Text to Columns" if necessary).
- Verify that your first row contains the headers: `TransactionID`, `Date`, `Branch`, `Product`, `Quantity`, `Price`.

Instructor Check

Ensure everyone has the dataset open and the headers are correctly aligned before proceeding.

Hands-on Exercise: Execution

1. **Sort by Date:** Arrange the "Date" column from Oldest to Newest. (Observe the timeline of sales).
2. **Sort by Price:** Arrange the "Price" column from Largest to Smallest. (Spot any unusually high or low prices).
3. **Filter by Branch:** Apply a filter to the "Branch" column and select only "Nairobi".
4. **Observe:** Look at the "Product" column for the Nairobi branch. Are there any spelling inconsistencies (e.g., "Laptop" vs "laptop")?

Takeaway

Notice how much easier it is to spot errors when the data is sorted and filtered!

Part 3: Removing Duplicates

"Ensuring every record is unique."

What are Duplicates?

- **Definition:** Rows in a dataset that contain identical (or nearly identical) information across key columns.
- **Types of Duplicates:**
  - **Exact Duplicates:** Every single column matches perfectly (e.g., the exact same transaction entered twice).
  - **Partial Duplicates:** Key identifiers match, but some details differ (e.g., the same Customer ID, but two different phone numbers).
- In ML, we generally focus on removing **Exact Duplicates** to prevent data inflation, while investigating **Partial Duplicates** to resolve conflicts.

How Duplicates Happen

- **Human Error:** A data entry clerk accidentally clicks "Submit" twice on a web form.
- **System Glitches:** A database sync fails and re-imports the same batch of records.
- **Merging Sources:** Combining two Excel files that both contain the same client list without checking for overlaps.
- **Lack of Validation:** The system allows the same email address or phone number to be registered multiple times.

The Danger

Duplicates artificially inflate the size of your dataset and give false weight to specific records, skewing the ML model's understanding of reality.

The Impact of Duplicates on ML Models

- **Overfitting:** The model memorizes the duplicated records, assuming they are highly important patterns, and fails to generalize to new data.
- **Biased Predictions:** If a specific customer profile is duplicated 50 times, the model will heavily bias its predictions toward that profile.
- **Inflated Metrics:** During testing, if duplicates leak into both the training and testing sets, the model's accuracy will appear artificially high (Data Leakage).

Removing duplicates ensures that every data point represents a unique, real-world event or entity.

Identifying Duplicates

Before removing duplicates, we must identify them.

Manual / Visual Identification

- **Sorting:** Sort by the primary key (e.g., "CustomerID" or "TransactionID"). Exact duplicates will now sit right next to each other.
- **Conditional Formatting (Excel):** Select the ID column → `Home > Conditional Formatting > Highlight Cells Rules > Duplicate Values`.

Highlighting duplicates visually allows you to review them before taking the destructive step of deletion.

Removing Duplicates in Microsoft Excel

Step-by-Step Guide

1. Select your entire dataset (including headers).
2. Go to the `Data` tab on the ribbon.
3. Click the `Remove Duplicates` button (in the Data Tools group).
4. A dialog box will appear. **Crucial Step:** Check "My data has headers".
5. Select the columns that define a unique record (usually all columns, or just the primary key like "TransactionID").
6. Click `OK`. Excel will report how many duplicate rows were removed.

Removing Duplicates in Google Sheets

Step-by-Step Guide

1. Select your entire dataset.
2. Go to the top menu: `Data > Data cleanup > Remove duplicates`.
3. A dialog box will appear. Check "Data has header row".
4. Choose to "Select all" or manually check the columns that define uniqueness.
5. Click `Remove duplicates`.
6. Sheets will show a confirmation toast at the bottom indicating how many rows were removed.

Safety First: Backing Up Before Deletion

The Golden Rule of Data Cleaning

NEVER overwrite your original raw data file!

- Always keep a master copy of the raw CSV (e.g., `messy_sales_data_RAW.csv`).
- Save your working file as a new version (e.g., `messy_sales_data_cleaning_v1.xlsx`).
- If you accidentally delete unique records during deduplication, you can always revert to the raw file and start over.
- This practice ensures your work is reproducible and auditable.

Hands-on Exercise: Removing Duplicates (Setup)

Preparation

- Return to your `messy_sales_data.csv` file.
- Ensure you have saved a backup copy of this raw file on your desktop.
- Identify the primary key column (e.g., `TransactionID`). This is the column that should theoretically be unique for every row.

Instructor Check

Confirm that everyone has identified the primary key and saved their raw backup.

Hands-on Exercise: Execution

1. **Identify:** Use Conditional Formatting (Excel) or simply sort by `TransactionID` to visually spot duplicate IDs.
2. **Remove:** Use the built-in "Remove Duplicates" tool.
   - Select all columns to ensure you only remove rows that are 100% identical across every field.
3. **Verify:** Check the row count before and after. Did the tool remove the expected number of duplicates?
4. **Save:** Save your progress. Your dataset is now free of exact duplicates!

Part 4: Handling Missing Values

"Deciding what to do with the gaps."

The Challenge of Missing Data

- In the real world, data is rarely complete. Users skip fields, sensors fail, and systems crash.
- **The ML Problem:** Most Machine Learning algorithms (like Linear Regression or Neural Networks) require a complete matrix of numbers. They cannot perform mathematical calculations on a "blank" space.
- If you feed missing data into a model, it will either throw an error and crash, or silently ignore the feature, reducing the model's predictive power.
- We must proactively decide how to handle these gaps before training.

How Missing Data Appears

Missing data is not always just an empty cell. Look out for these inconsistent placeholders:

- **True Blanks:** The cell is completely empty.
- **Text Placeholders:** "N/A", "NA", "Null", "None", "Unknown".
- **Symbol Placeholders:** "-", "?", "0" (when 0 is not a valid logical value, e.g., Age $= 0$).
- **Whitespace:** A cell that looks empty but actually contains a space character (" ").

The Danger of Placeholders

If a column is supposed to be numerical, but contains the text "N/A", the entire column will be treated as text, breaking your ML model.

Why Missing Data is a Problem for ML

- **Loss of Information:** If you have to drop rows with missing data, you lose valuable historical records, reducing your training set size.
- **Introduction of Bias:** If data is missing *systematically* (e.g., high-income users refuse to share their age), filling the gaps with the average age will create a biased, inaccurate model.
- **Algorithm Failure:** As mentioned, math operations cannot be performed on null values.

The goal is to handle missing data in a way that preserves the true underlying distribution of the dataset.

Identifying Missing Data in Spreadsheets

Method 1: Filtering

Apply a filter to a column and uncheck all values except "Blanks" (or search for "N/A").

Method 2: Go To Special (Excel)

Press `F5` > `Special` > `Blanks`. This will highlight every empty cell in your selection.

Method 3: COUNTIF Formula

Use `=COUNTIF(A:A, "")` to count the exact number of blank cells in a column.

Identifying the volume of missing data helps you decide which strategy (deletion vs. imputation) to use.

Strategies for Missing Data: Overview

There are two primary strategies for handling missing values:

1. Deletion (Dropping)

- Remove the incomplete row or column entirely.
- *Pros:* Simple, no assumptions made.
- *Cons:* Loss of data.

2. Imputation (Filling)

- Fill the gap with a calculated or logical value.
- *Pros:* Retains all rows.
- *Cons:* Introduces statistical assumptions.

The choice depends on *how much* data is missing and *why* it is missing.

Strategy 1: Deletion (Dropping)

- **Dropping Rows:** If a single row is missing a critical value (e.g., the "Target" variable is blank), you simply delete that entire row.
- **Dropping Columns:** If an entire column is mostly empty (e.g., 80% of the "MiddleName" field is blank), it provides no value to the ML model. Delete the entire column.

When to use Deletion

Use deletion when the missing data is completely random, affects only a tiny fraction of your dataset (e.g., $< 5\%$), or when the missing column is irrelevant to the prediction goal.

When to Use Deletion

Scenario A: Drop the Row

You are predicting "Customer Churn". A row is missing the "Churned" status (the target variable). You cannot train the model without knowing the outcome, so you **delete the row**.

Scenario B: Drop the Column

You are predicting "Sales". The dataset includes a "Fax Number" column, but 90% of customers don't have fax numbers. The column is mostly blank and irrelevant, so you **delete the column**.

Strategy 2: Imputation (Filling the Gaps)

- **Definition:** Replacing missing data with a substituted value based on the rest of the dataset.
- **Why Impute?** It allows you to keep the row in the dataset, preserving the information in the other columns.
- **Common Imputation Techniques:**
  - **Mean (Average):** For numerical data with a normal distribution (e.g., filling missing "Salary" with the average salary).
  - **Median (Middle value):** For numerical data with outliers/skewed distribution (e.g., filling missing "Age").
  - **Mode (Most frequent):** For categorical data (e.g., filling missing "City" with "Nairobi" because it's the most common).

Imputation Techniques in Spreadsheets

How to Impute in Excel/Sheets

1. Calculate the statistic in an empty cell (e.g., `=MEDIAN(B2:B1000)`).
2. Copy that calculated value.
3. Filter the column to show only "Blanks".
4. Select the blank cells, paste the calculated value, and clear the filter.

Advanced Imputation

You can also fill missing values with a specific placeholder like "Unknown" for categorical data. This allows the ML model to learn that "Unknown" is its own distinct category.

Day 2 Assignment

Task: The Data Cleaning Checklist

- Complete the "Data Cleaning Checklist" worksheet provided by the instructor.
- Note **every tool/button** used to clean the Friday dataset.
- **Reflect:** Write a brief justification for the missing value strategy you chose for the "Age" column (e.g., "I chose Median over Mean because there were extreme outliers in the age data").

Submission

Upload the completed checklist and your final cleaned dataset (`cleaned_sales_data.csv`) to the Mediacrest portal before Saturday's session.

Q&A and Wrap-up

Questions?

"Clean data is the foundation of powerful AI."

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